ROSHAN L

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Education

Alliance University, Bengaluru **AUG 2023 – MAY 2027**
B.Tech in Information Technology (Data Analytics) *Anekal ,Karnataka*

Relevant Coursework

- Data Structures & Algorithms
- Database Management Systems
- Probability & Statistics
- Machine Learning
- Artificial Intelligence
- Operating Systems
- Data Analysis
- Computer Architecture

Experience

White and Box – Tech Products & Services **Dec 2025 – Mar 2026**
Data Science Intern *HSR Layout, Karnataka*

- Cleaned and preprocessed datasets using Pandas and NumPy .
- Performed exploratory data analysis (EDA) to identify trends and patterns.
- Handled missing values, duplicates, and data transformation tasks .
- Created interactive Tableau dashboards and visual reports .
- Presented findings using dashboards and data storytelling techniques

Projects

Skin Lesion Detection using Deep Learning| *TensorFlow, CNN* **February 2026**

- Built end-to-end AI pipeline from raw image → diagnosis → report
- Enabled 3D lesion visualization, improving interpretability
- Combined deep learning + classical CV techniques for robustness and designed real-time interactive web interface.
- Implemented explainable AI (Grad-CAM) for trust in predictions

Offline AI Voice Keyboard System| *Python, Vosk, PyQt5, LanguageTool (Ubuntu/Linux)* **November 2025**

- Developed a fully offline AI pipeline on Ubuntu/Linux, converting speech → text → grammar-corrected output → system-wide text insertion
- Integrated Vosk speech recognition model for real-time, low-latency voice-to-text processing without internet dependency.
- Implemented grammar correction using LanguageTool (local NLP server) to enhance text accuracy and readability.
- Designed a floating, interactive UI using PyQt5 with mic control, keyboard access, and accessibility-focused features.
- Built Linux system-level automation using xdotool and PyAutoGUI to detect active windows and inject text into any application

AI-Powered Driver Safety System (Smartwatch) | *Python, ML, IoT* **August 2025**

- Built real-time ML-based system for detecting driver distress using sensor data .
- Integrated GPS and BLE communication for real-time monitoring and alerts.
- Designed system for low-latency edge inference.
- Implemented automated emergency alert mechanism.

Technical Skills

Python, SQL | Pandas, NumPy | ML (Scikit-learn, TensorFlow, Model Evaluation)
Data Preprocessing, EDA | Tableau, Matplotlib | CNN | Git

Achievements

- Winner – Best Paper Presentation (NCSS AI'26 Conference)
- Winner – Best Use of Open Source (National Level Open-Source Hackathon, REVA University)
- Participated – PRAKALP 2025 Science Expo

Publications

Enhanced Skin Lesion Diagnosis Through Real-Time Deep Learning and Transfer Learning
• International Journal for Multidisciplinary Research (IJFMR), Vol 8, Issue 2, 2026